GOVERNMENT OF WESTERN AUSTRALIA

GSWA Geochronology Record

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Department of Mines, Industry Regulation and Safety

211118.1.pdf *Geochronology Record 1930*

211118.1: lamprophyre dyke, Deflector prospect

(Yilgarn Craton)

SS Romano, MTD Wingate, Y LU, IOH Fielding and TJ Ivanic

Location and sampling

YALGOO SH50-02, MELLENBYE 2140 Zone 50, MGA 439143E 6828066N
Warox Site TJIBYE000006 Sampled on 20 May 2019

This sample was collected from the 174.0 – 174.5 m depth interval of diamond drillcore BDD018, drilled in 2006 by Batavia Mining Ltd at their Deflector gold–copper prospect (Mattinson, 2007). The drillhole is located about 8.0 km west-northwest of Buddadoo Well, 7.5 km east-northeast of Yallabyne Well, and 0.3 km south of the Deflector West open pit.

Geological context

The unit sampled is a metamorphosed lamprophyre dyke within the Gullewa greenstone belt of the Yilgarn Craton. The dyke is intruded into metamorphosed volcanogenic and sedimentary rocks of the 2800–2735 Ma Polelle Group (Mattinson, 2007; Van Kranendonk et al., 2013). A minette dyke with sharp margins was identified at a depth of 109.9 m in this drillcore (sample 42266, Appendix 1, Bishop, 2007). An altered polymictic breccia of the Mougooderra Formation (Zibra, 2022: ENS report Mougooderra Formation, viewed 2 August 2023), sampled in drillcore BDRCD075 about 0.2 km to the northwest, yielded 3087–2625 Ma detrital zircons and a conservative maximum depositional age of 2672 ± 10 Ma (GSWA 211120, Wingate et al., 2021).

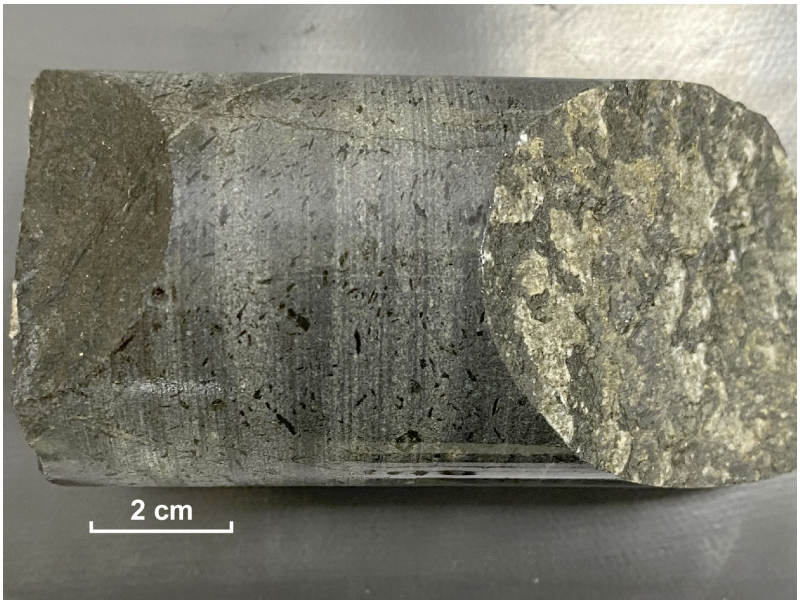

Figure 1. Drillcore image for sample 211118: lamprophyre dyke, Deflector prospect

Petrographic description

The sample is a fine to medium-grained lamprophyre, consisting of about 60% feldspar, 18% biotite, 12% actinolite, 5% quartz, 4% magnetite, 1% titanite, trace hornblende and accessory apatite. Secondary minerals include quartz, carbonate and trace pyrite and chalcopyrite. Feldspar is anhedral to tabular, up to 0.7 mm long, and forms aggregates of randomly oriented laths. Based on twinning, K-feldspar is more abundant than plagioclase, which is albitized, and all feldspars are overprinted by fine-grained biotite, amphibole and magnetite. A trace amount of granophyre is present within the feldspar aggregates. Brown biotite is anhedral to euhedral and forms randomly oriented grains up to 2 mm long, as well as fine-grained aggregates around larger biotite laths, in feldspar-rich domains, and along fractures. Pale green actinolite occurs as ragged relicts, up to 1 mm in size, of original subhedral laths, and as dispersed grains up to 0.1 mm in size. Quartz is

interstitial, anhedral, irregular, up to 0.25 mm in size, and appears to be mostly secondary. Magnetite forms anhedral to subhedral, equant grains up to 0.3 mm in diameter, and also disseminated micron-scale grains. Irregular, granular masses of titanite are up to 0.7 mm in size and associated with actinolite. Tabular subhedral to euhedral apatite is up to 0.7 mm long. Hornblende is subhedral and up to 0.5 mm. Pyrite forms cubic grains up to 0.5 mm across and anhedral chalcopyrite is associated with actinolite. The sample lacks obvious features of deformation, although fine-grained biotite and actinolite are interpreted to be products of contact metamorphism at biotite to hornblende hornfels grade.

Zircon morphology

Zircons isolated from this sample are colourless and anhedral to subhedral. The crystals are up to 120 µm long, and equant, with aspect ratios up to 2:1. In cathodoluminescence (CL) images all zircons exhibit concentric zoning. A CL image of all zircons is shown in Figure 2.

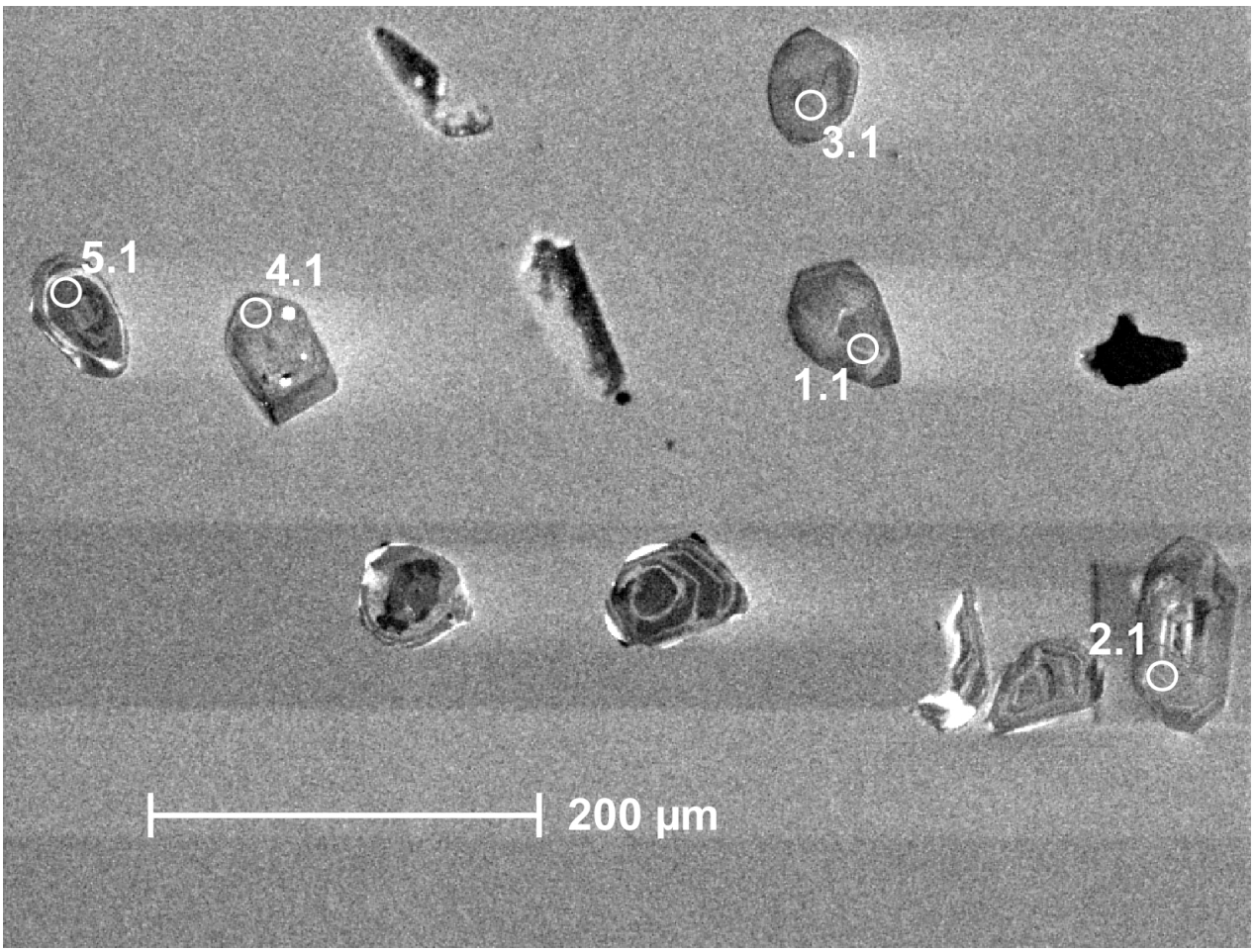

Figure 2. Cathodoluminescence image of all zircons from sample 211118: lamprophyre dyke, Deflector prospect. Numbered circles indicate the approximate locations of analysis sites

Analytical details

This sample was analysed on 21–22 October 2021, using SHRIMP-B. Fifteen analyses of the M257 standard were obtained during the session, of which 11 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.80% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.32% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1 (provided in the Links section). Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Five analyses were obtained from five zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to moderately discordant (Fig. 3). One analysis is >5% discordant. The date obtained from this analysis (Table 1, Group D) is unreliable, and considered not to be geologically significant. The remaining four analyses form a single dispersed group.

Group X comprises four analyses (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 2871–2624 Ma.

The dates of 2871–2624 Ma for the four analyses in Group X are interpreted as the ages of inherited zircons. The date of 2624 ± 13 Ma (1σ) for the youngest zircon in Group X is tentatively interpreted as a maximum age for intrusion and igneous crystallization of the lamprophyre dyke.

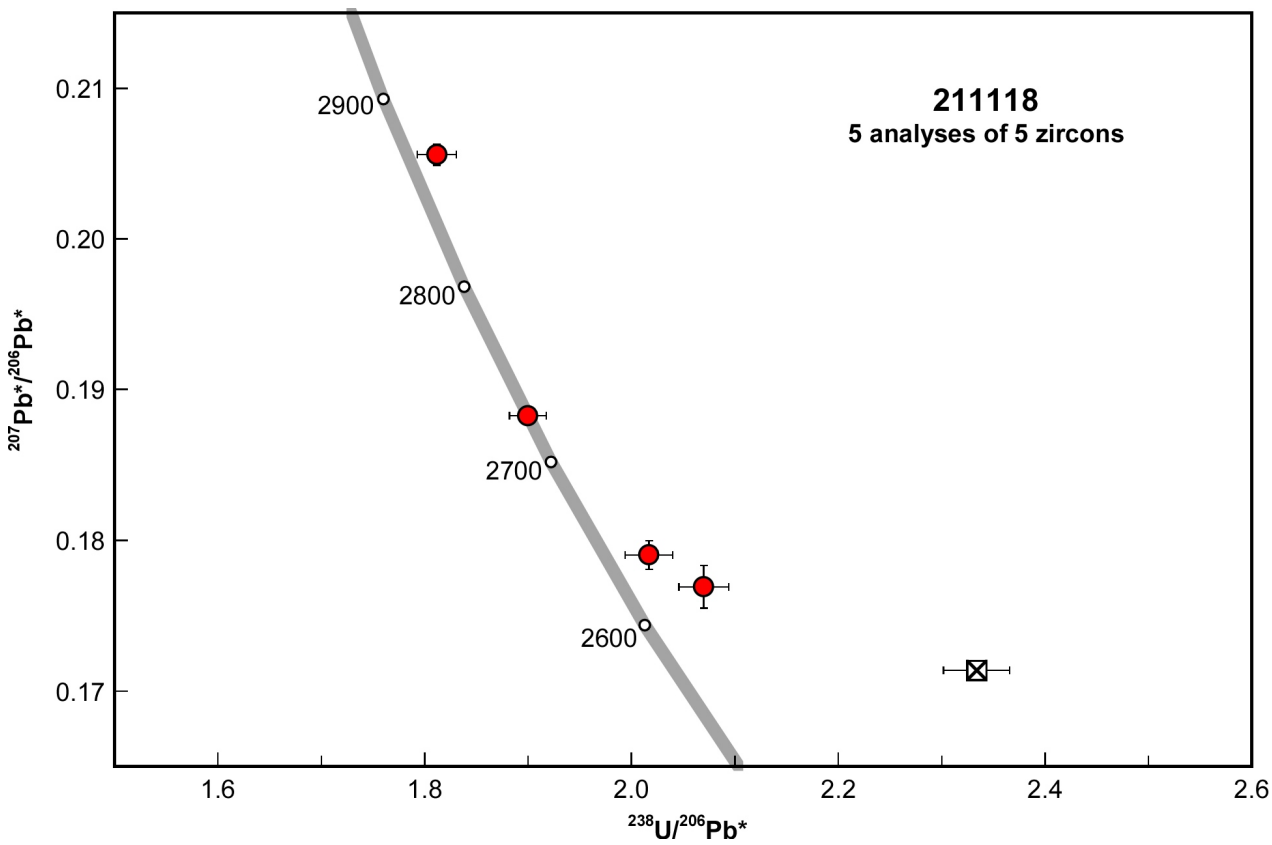

Figure 3. **U–Pb analytical data for zircons from sample 211118: lamprophyre dyke, Deflector prospect. Red circles indicate Group X (xenocrystic zircons); crossed square indicates Group D (discordance >5%)**

Links

Table 1: Ion microprobe analytical results for zircons from sample 211118: lamprophyre dyke, Deflector prospect: 211118.1.Table1.xls
SHRIMP spot images: 211118_spots.pdf
Introduction to geochronology information: Intro_2023.pdf

References

Bishop, J 2007, Batavia Mining Ltd, Annual Technical Report for the year ending 31 December 2007, Gullewa Project C15/1997; Department of Mines, Industry Regulation and Safety; WAMEX Report A77005: Geological Survey of Western Australia, 30p.

Mattinson, P 2007, Batavia Mining Ltd, Annual Technical Report, Gullewa Project, 1 January 2006 to 31 December 2006; Department of Mines, Industry Regulation and Safety; WAMEX Report A74115: Geological Survey of Western Australia, 31p.

Stacey, JS and Kramers, JD 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221.

Van Kranendonk, MJ, Ivanic, TJ, Wingate, MTD, Kirkland, CL and Wyche, S 2013, Long-lived, autochthonous development of the Archean Murchison Domain, and implications for Yilgarn Craton tectonics: Precambrian Research, v. 229, p. 49–92.

Wingate, MTD, Lu, Y, Fielding, IOH and Ivanic, TJ 2021, 211120: altered breccia, Deflector prospect; Geochronology Record 1823: Geological Survey of Western Australia, <www.dmirs.wa.gov.au/geochron>. View Reference online

Recommended reference for this publication

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extract viewed 13 December 2023. <www.dmirs.wa.gov.au/geochron>

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Grid references in this publication refer to the Geocentric Datum of Australia 2020 (GDA20). All locations are quoted to at least the nearest 100 m.

Capitalized names in text refer to standard 1:100 000 map sheets, unless otherwise indicated.

WAROX is GSWA's field observation and sample database. WAROX site IDs have the format 'ABCXXXnnnnnnSS', where ABC = geologist username, XXX = project or map code, nnnnnn = 6-digit site number, and SS = optional alphabetic suffix (maximum 2 characters).

Isotope and element analyses are routinely conducted using the GeoHistory laser ablation ICP-MS and Sensitive High-Resolution Ion Microprobe (SHRIMP) facilities at the John de Laeter Centre (JdLC), Curtin University, with the financial support of the Australian Research Council and AuScope National Collaborative Research Infrastructure Strategy (NCRIS), and the Microscopy Australia laser ablation ICP-MS facility at the Centre for Microscopy, Characterisation & Analysis (CMCA), The University of Western Australia (UWA). The TESCAN Integrated Mineral Analyser (TIMA) instrument was funded by a grant from the Australian Research Council (LE140100150) and is operated by the JdLC with the support of the Geological Survey of Western Australia, UWA and Murdoch University. Mineral analyses are routinely obtained using the electron probe microanalyser (EPMA) facilities at the CMCA at UWA, at Adelaide Microscopy at the University of Adelaide and the Central Science Laboratory at the University of Tasmania.

Digital data related to WA Geology Online, including geochronology and digital geology, are available online at the Department's Data and Software Centre and may be viewed in map context at GeoVIEW.WA. For further information about this Geochronology Record, please contact geochronology@dmirs.wa.gov.au and quote the geochronology record and/or the sample number.

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Acknowledgment of Country

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